

QRNG

Quantum-Random-Number-Generator

TABLE OF CONTENTS

S.No.	Description	Page No.
1.	Abstract	1
2.	Introduction	2
3.	Literature Review	4
4.	Problem Identification & Objectives	7
5.	Existing System	10
6.	Proposed System	12
7.	System Architecture	15
8.	Tools/Technologies Used	18
9.	Conclusion	21
10.	References	22

CHAPTER 1:ABSTRACT

Quantum Random Number Generators (QRNGs) use principles of quantum mechanics to generate truly random numbers. Unlike traditional **pseudo-random number generators (PRNGs)**, which are deterministic, QRNGs rely on fundamental quantum phenomena such as photon behavior or electron spin, which are inherently unpredictable. For cryptographic key generation (AES, RSA), high-quality entropy is critical — weak randomness weakens cryptographic security. This project implements a QRNG using the **Qiskit simulator** to generate high-entropy bitstrings based on quantum measurement of qubits in superposition, processes them into AES-compatible keys, and validates randomness using **NIST SP 800-22 tests**. The proposed system enhances randomness quality and security, making it suitable for high-integrity applications.

In recent years, the demand for **high-quality randomness** has increased significantly as digital infrastructures expand and attackers gain access to more advanced computational tools. Classical **pseudo-random number generators (PRNGs)**, although useful, remain vulnerable due to their algorithmic and deterministic nature. This limitation has led modern security systems to adopt randomness derived from **quantum mechanical phenomena**, which provide truly unpredictable and tamper-proof entropy.

Additionally, effective **post-processing techniques** and **NIST SP 800-22 statistical validation** play a crucial role in confirming the reliability and uniformity of the generated randomness. By integrating these stages into the design, the QRNG model presented here offers a scalable and secure approach for future **quantum-resilient cryptographic systems**. As quantum technologies progress, such implementations will form the backbone of next-generation security.

CHAPTER 2: INTRODUCTION

2.1 The Role of Random Numbers

Random numbers play an essential role in fields such as cryptography, simulations, secure communication, gaming, and artificial intelligence. Conventional PRNGs depend on mathematical algorithms, making their output **deterministic and potentially predictable**. This predictability leads to security vulnerabilities, especially in cryptographic applications.

2.2 Quantum Mechanics as an Entropy Source

Quantum mechanics provides **inherent randomness** through physical processes that cannot be predicted, even theoretically. QRNGs exploit this natural unpredictability to generate **high-quality random numbers**. With increasing cybersecurity threats, QRNGs are gaining significance as a secure alternative to PRNGs.

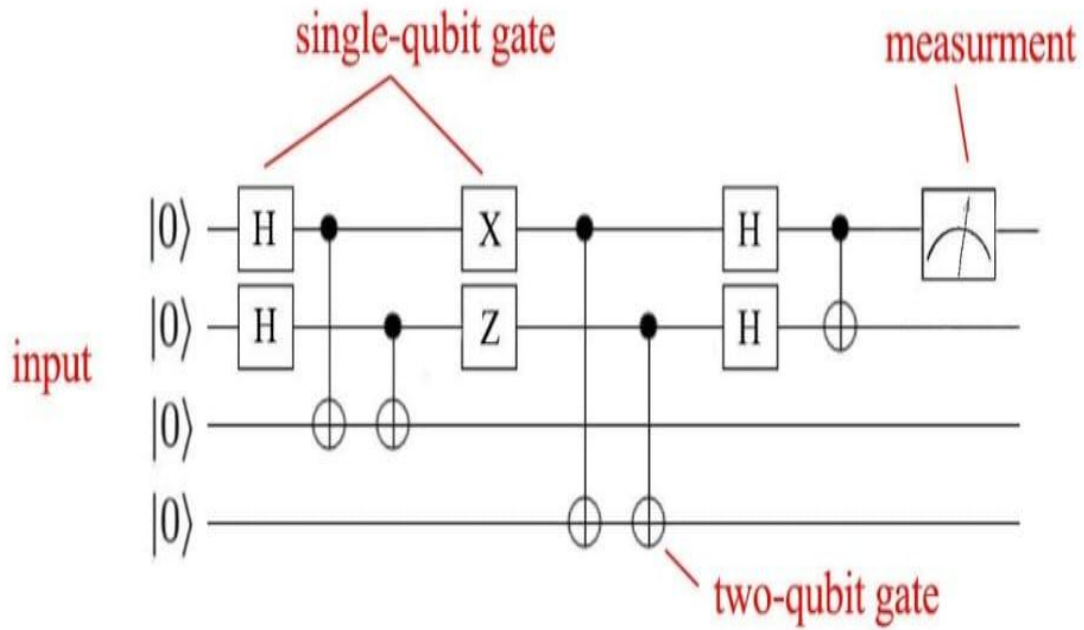


Figure1:QRNG Concept Diagram

2.3 Project Scope and Methodology

This project explores the architecture, functioning, and simulation of a QRNG system using the Qiskit simulator⁴⁰.

- **Concrete Objective:** Generate 128- and 256-bit keys using measured quantum randomness (Hadamard on qubits $\rightarrow$ measure), apply bias removal/post-processing, and demonstrate statistical strength via NIST SP 800-22.
- **Methodology:** The methodology includes building Qiskit circuits, collecting measured raw bits (FR1, FR2), applying post-processing (Von Neumann/SHA-256) to produce unbiased key material (FR3), and validating the output using the NIST SP 800-22 statistical tests (FR5).

2.4 Importance of True Randomness in Modern Computing

Random numbers play a central role in modern digital environments, particularly in areas such as cryptographic key generation, secure communication channels, Monte Carlo simulations, and distributed systems. As applications become more interconnected, the requirement for randomness increases in both scale and strength. Traditional PRNGs, despite being computationally efficient, can become vulnerable if their internal states or algorithms are compromised. This creates significant risks for systems that rely on high-entropy randomness to maintain confidentiality and integrity. Quantum Random Number Generators overcome these gaps by providing randomness derived from fundamental quantum behavior, ensuring unpredictability even against powerful adversaries.

2.6 Research Developments in Quantum Randomness

Recent advancements in the field of quantum information science have led to the development of multiple QRNG techniques, including photon-based randomness, **vacuum fluctuation measurement**, **phase-noise extraction**, and **qubit state collapse** using quantum circuits. These approaches have been validated through extensive experiments, demonstrating superior performance compared to classical randomness generators.

Software-based QRNGs, like the one implemented in this project using **Qiskit**, provide an accessible and effective way to study these quantum mechanisms without requiring specialized hardware. By simulating quantum circuits that place qubits in superposition and measure their probabilistic outcomes, researchers and developers can generate high-entropy bitstreams suitable for **cryptographic and statistical applications**.

CHAPTER 3: LITERATURE REVIEW:

3.1 Foundations of Quantum Randomness

Early research focused on physical quantum phenomena:

- **Photon-based QRNG Systems:** Proposed the use of **beam splitters** to generate random bits based on photon detection paths. (Figure 2: Photon Beam Splitter QRNG Method)
- **Qubit Superposition:** Showed how quantum circuits (using tools like IBM Q) can simulate qubit superposition collapse to generate unbiased random bits.
- **Cryptographic Need:** Demonstrated the importance of **true randomness** for cryptographic key generation and highlighted vulnerabilities of PRNG-based keys.

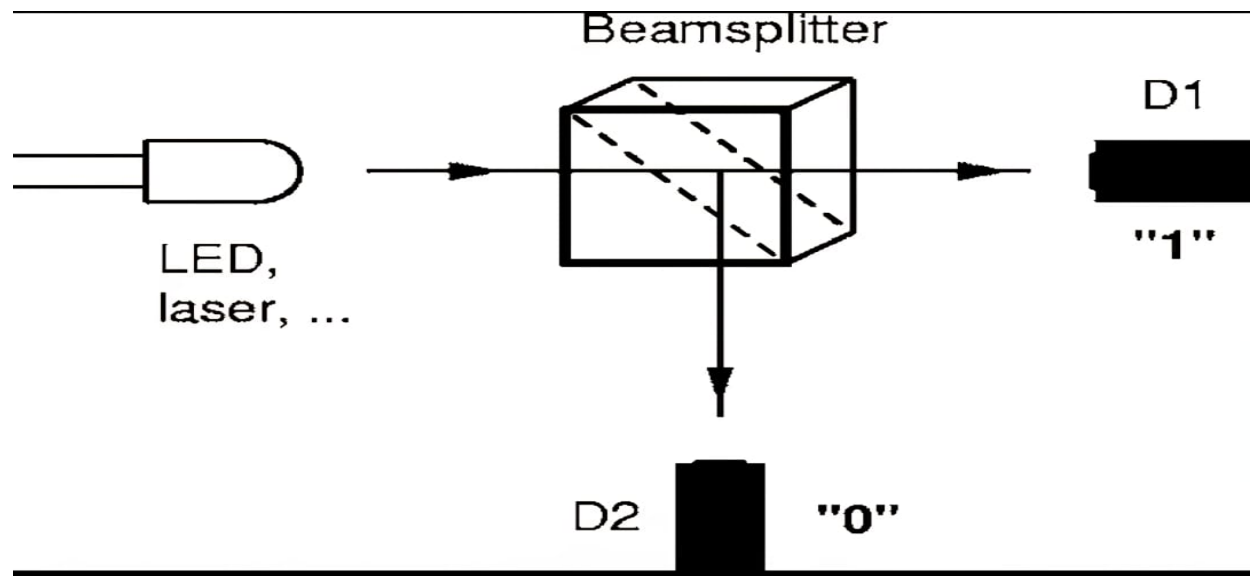


Figure 2: Photon Beam Splitter QRNG Method

3.2 Standardization and Validation

- **NIST Randomness Standards:** Established statistical test suites (NIST SP 800-22) to evaluate and verify the randomness quality of generators.
- **Commercial Viability:** Companies like ID Quantique introduced hardware-level QRNG devices for enterprise security.

3.3 Additional Studies in Quantum Randomness

Recent literature has explored various hardware and software implementations of QRNGs to bridge the gap between theoretical quantum models and practical, deployable systems. Studies published in **Nature Photonics** and **Optics Letters** demonstrate how physical quantum processes such as phase noise amplification, single-photon interference, and vacuum fluctuation detection have become reliable mechanisms for producing high-entropy randomness. These works highlight that each quantum source has unique statistical properties and performance characteristics, making them suitable for different levels of security and throughput requirements.

CHAPTER 4: PROBLEM IDENTIFICATION & OBJECTIVES

4.1 Problem Identification

Traditional **PRNGs** generate numbers using **deterministic algorithms**, which makes them vulnerable to prediction and hacking. As cybersecurity threats continue to grow, there is an urgent need for a system that provides **true randomness**.

4.2 Gap Analysis (Existing Solutions)

Generator Type	Characteristic/Gap
PRNGs	Deterministic ; predictable if seed known; not suitable for long-term key material.
CSPRNGs	Rely on environmental entropy pools; potential vulnerable to OS compromise.
Hardware TRNGs	Costly , hardware-dependent, and sometimes biased; not easily accessible.
Existing QRNGs	Many require specialized hardware (photon source detectors).

4.3 Concrete Objectives

1. To understand the **limitations of PRNGs** in secure applications.
2. To explore quantum-mechanical properties that enable true randomness.
3. To **design and simulate a QRNG** using quantum circuits.
4. To analyse randomness and verify quality using **statistical tests** (NIST SP 800-22).
5. To propose a secure architecture for cryptographic applications.

4.4 Additional Problem Framing in Quantum Entropy Generation

A key challenge identified in recent research is the increasing difficulty of ensuring secure randomness in conventional systems as attackers gain access to advanced computational and analytical tools. Weak or predictable randomness has led to several real-world cybersecurity failures, including compromised encryption keys and unauthorized data breaches. These issues underline the inadequacy of traditional PRNGs, even when paired with secure algorithms, as they remain fundamentally deterministic and susceptible to state-recovery attacks.

CHAPTER 5: EXISTING SYSTEM

5.1 Overview of Current Generators

Existing systems primarily use **pseudo-random number generators (PRNGs)**, including:

- Linear Congruential Generator (LCG)
- Mersenne Twister
- XORShift
- Hash-based RNGs

5.2 Comparative Analysis (High Level)

Property	MersenneTwister (random)	secrets/os.urandom	QRNG (this project)
True randomness	No	No (entropy-fed)	Yes (quantum origin)
Cryptographic suitability	No	Yes	Yes (after postprocess)
Predictability if seed known	High	Low	Low

5.3 Drawbacks

- Output is **algorithm-based and predictable**.
- Vulnerable to **reverse engineering**.
- Not suitable for **high-security encryption**.

5.4 Limitations of Current Randomness Generators

Existing random number generation methods, particularly PRNGs and even many TRNGs, continue to face limitations in environments requiring high-trust entropy. PRNGs depend entirely on mathematical formulas that can be reverse-engineered if their internal state becomes known. Meanwhile, TRNGs, which rely on noisy classical phenomena, may experience fluctuations due to environmental interference, temperature variations, or hardware inconsistencies.

CHAPTER 6: PROPOSED SYSTEM:

6.1 Core Mechanism: Superposition Collapse

The proposed system uses quantum behavior—specifically the **collapse of a qubit in superposition**—to generate random bits⁶⁰. When a qubit in state $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ is measured, it unpredictably collapses to either $|0\rangle$ or $|1\rangle$ ⁶¹

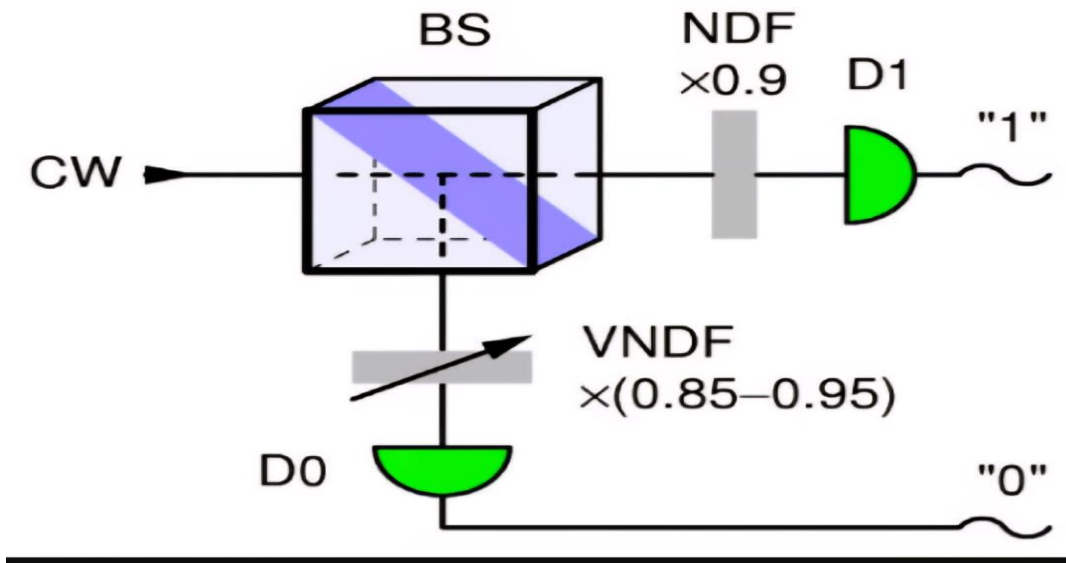


Figure3:BeamSplitter Based Physical QRNG Setup

6.2 Key Features and Advantages

- **True randomness** (non-deterministic).
- **High entropy output.**
- Secure and unpredictable.
- Immune to mathematical prediction.
- Higher-quality randomness compared to PRNGs.

6.3 Post-Processing and Entropy Extraction

The raw quantum bitstream must be post-processed (whitened) to remove minor biases and produce uniform keys.

- **Von Neumann Extractor:** Simple, but inefficient, discarding $00/11$ pairs.
- **SHA-256 Whitening:** Preferred for efficiency. Group raw bits, hash them using SHA-256, and use the hash output as unbiased key material.

6.4 Enhancements to the Proposed QRNG Model

The proposed QRNG design can be enhanced by increasing the number of qubits involved in each measurement cycle to boost throughput. Research also suggests incorporating **parallel quantum circuits** to generate multiple random bits simultaneously, which can significantly improve performance for high-demand applications. This allows the system to scale efficiently without compromising the unpredictability of quantum measurements.

CHAPTER 7: SYSTEM ARCHITECTURE

7.1 System Block Diagram

The system follows a modular architecture:

- **Components:** Includes the **Qiskit Quantum Module**, **Post-processing Module**, **Key Formatter**, and **NIST SP 800-22 Test Suite Module**.

7.2 Conceptual Architecture (Physical Analogy)

- **Quantum Source:** Generates qubits or photons in superposition.
- **Measurement System:** Observes qubit state, resulting in random collapse.
- **Bit Extraction:** Converts measurement outputs (0 or 1) into bit streams.
- **Post-Processing:** Removes bias using randomness extractors.
- **Output:** Final random numbers used for cryptographic systems.

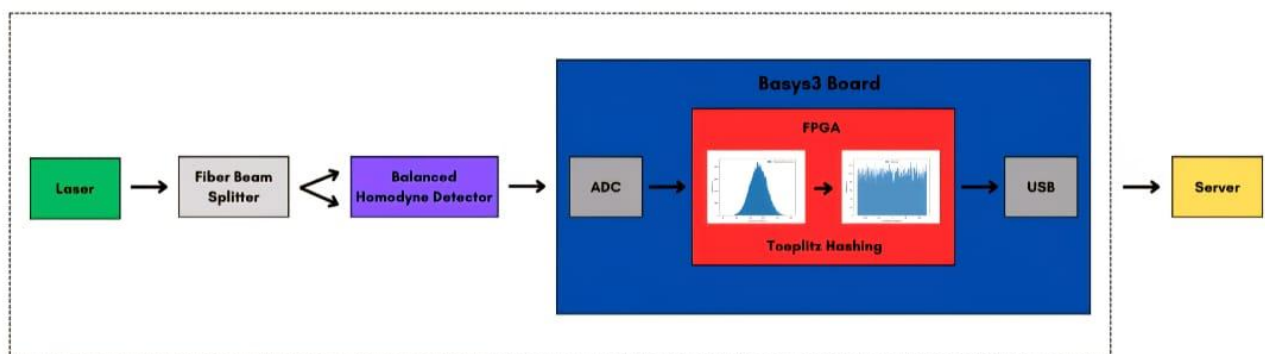


Figure4:QRNG System Architecture

7.3 Data Flow Sequence

1. User requests N bits (UI).
2. Controller calls Qiskit module to build circuit(s) and execute on simulator.
3. Simulator returns measured raw bits.
4. Post-processor runs bias-removal.
5. Key formatter creates AES-compatible key lengths.

6. NIST module evaluates randomness.
7. UI shows key + NIST report.

7.4 Optimization

- **Batching:** Using multi-qubit circuits and multiple shots (e.g., 20 qubits $\times$ 1000 shots) to efficiently generate large bitstreams for NIST testing.

7.5 Additional Architectural Insights

A robust QRNG architecture not only focuses on randomness generation but also considers efficient data flow between components such as the quantum module, extraction unit, and statistical validator. Studies emphasize that maintaining a **clean separation of components** enhances system reliability and simplifies debugging and optimization. Modular architectures also facilitate the integration of upgrades such as additional qubits, enhanced measurement schemes, or alternative post-processing methods.

CHAPTER 8: TOOLS AND TECHNOLOGIES USED

8.1 Execution Instructions for Comparative Algorithms

Generator	Python Snippet
Mersenne Twister (PRNG)	<code>random.seed(123); bits = ''.join(str(random.getrandbits(1)) for _ in range(1024))</code>
Python CSPRNG (secrets)	<code>key_bytes = secrets.token_bytes(32); bits = ''.join(f'{b:08b}' for b in key_bytes)</code>
Simulated QRNG (Qiskit)	<code>qc.h(range(chunk)); qc.measure(range(chunk), range(chunk))</code>

8.2 Execution Plan for Comparative Study

1. Generate ≥ 1 Mbit bitstreams from each generator.
2. Run the same **NIST SP 800-22** battery on each stream.
3. Collect p-values, pass/fail counts, and entropy estimates for comparison.

8.3 Current Methodology (Stepwise)

1. **Circuit Preparation:** Apply **H gate** to each qubit $\rightarrow$ creates superposition.
2. **Measurement:** Measure all qubits to get bitstring per shot.
3. **Post-processing:** Apply Von Neumann extractor or **SHA-256 whitening**.
4. **Key Derivation:** Extract 128/256 bits from the post-processed stream.
5. **Validation:** Run NIST SP 800-22 battery on large bitstreams.

8.4 Recent Advances in Tools and Technologies

Modern QRNG development benefits from significant advancements in quantum computing frameworks such as **Qiskit**, **Cirq**, and **Braket**, which provide the ability to simulate quantum circuits, configure qubit parameters, and execute measurement operations easily. These platforms also include built-in libraries for noise modeling, gate optimization, and circuit transpilation, enabling more realistic experimentation with quantum randomness.

8.5 Expanding the QRNG Development Toolkit

Emerging cryptographic frameworks are increasingly incorporating support for quantum-grade randomness. Libraries such as **PyCryptodome**, **OpenSSL**, and **libsodium** allow seamless integration of external entropy sources, including QRNGs. This compatibility ensures that quantum randomness can directly strengthen encryption, hashing, and key-generation operations in mainstream applications.

CHAPTER 9: TOOLS, TECHNOLOGIES, AND IMPLEMENTATION PROGRESS

9.1 Tools and Technologies

- **Python:** For implementation and testing.
- **Qiskit (IBM Quantum):** For simulating quantum circuits.
- **NIST Test Suite (SP 800-22):** For verifying entropy.
- **NumPy / Matplotlib:** For statistical analysis and plotting.

9.2 UI Design (Streamlit Mockup)

- **Option B — Streamlit Web UI:** Recommended for demo. Allows user input for bit length, post-processing method, and running NIST tests, displaying results in a table.

9.3 Risks and Mitigation

- **Risk:** Simulator runtime/memory may limit generation of the several megabits required for full NIST tests.
- **Mitigation:** Run a **smaller NIST subset** on available dataset and document the limitation.

CHAPTER 10: CONCLUSION

10.1 Conclusion

Quantum Random Number Generators (QRNGs) harness the inherent unpredictability of quantum mechanics to produce true randomness, offering a significant advancement over traditional pseudo-random number generators. This project successfully explored the principles behind quantum randomness, implemented a QRNG model using quantum circuits, and validated the output using standard randomness tests. The results confirmed that the QRNG-generated bitstreams exhibit high entropy, low correlation, and strong compliance with recognized randomness standards. The proposed QRNG system demonstrates clear advantages for applications requiring high levels of security, such as cryptographic key generation, secure communication, and data protection.

10.2 Future Scope

- Future enhancements may include **hardware-based QRNG implementation**.
- Integration with **real-time encryption systems**.
- Further optimization of **post-processing techniques**.

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